PROJECT DIRECTIVE · v0.2

CONCURRENT

Risk II SameTime. Chassis update. Not a new genre.

v0.1 locked the right soul and the right build order, then failed its own combat brief and left holes in rules that were marked locked.

v0.2 does not add missions, factions, or a clever new hook.
The hook is SameTime: everyone commits, attackers leave home empty, the map detonates, the person who stripped a border gets walked.
2026 is cinema, phones as order pads, ghost arrows, match replays, and combat a stranger can parse without a spreadsheet.

THIS REVISION

COMBAT	Revoked and replaced. Size + 1d6. One roll. Bands hold.
TACTICAL	Folded into the next sealed sheet. One commit per player.
NET / BOT	Random-legal bot in week 1. Phone-pad room with the 7-map.
WIN	Score after N rounds default. Domination is an early end.
IP	Still not Risk. Original names. Same density of decision, not the same board.

19 Sep 2026 · Amends v0.1 · Combat 3.5 is dead

00 — WHAT CHANGED AND WHAT DID NOT

The game is still Risk II SameTime. Simultaneous plans. Attackers vacate. Multiple forces can hit one hex. Restricted placement. Surge last. That is the wheel. v0.1 was right about the wheel and about the build order. It was wrong about combat, sloppy about several “locked” rules, and it scheduled hotseat and a late bot in ways that fight the thesis.

Do not add a secret-mission layer to make it “modern.” Modern is: one device per player, a resolution cinema that makes the detonation visible, ghost arrows from last round, exact replay files, and a garrison cap readout so hidden reinforcements have a known ceiling. SameTime already had the idea. 2026 has the tools.

01 — THE WHEEL (DO NOT RENEGOTIATE)

If a later idea fights this list, the idea loses.

- **Simultaneous commit.** Reinforcements, attacks, and the tactical/regroup are sealed before anything resolves.
- **Attackers vacate.** Strength assigned to an attack is gone from the origin when orders lock. Origins defend with what is left. Never soften this.
- **Convergence.** Several packets may legally target the same destination. Different owners do not merge. Same owner does.
- **Fixed queue.** Border clashes → mass / single invasions → spoils → surge. No click-order.
- **Placement cap.** Armies added to T this round ≤ 1 + friendly neighbors of T. Unused income is lost.
- **Hidden intent, public map.** You see army counts. You do not see sealed orders. Show each enemy hex’s *max possible garrison after reinforce* (current + cap).

02 — PRODUCT SHAPE

Digital-first board game. Original title, original map names, original region names. Graph density may rhyme with classic conquest maps. The product is not a licensed clone.

Layer	Ship	Gate
L0 Kernel	Pure TS. Deterministic. No DOM.	T1–T8 plus property tests pass.
L1 Harness	CLI, JSON fixtures, combat_sim, replays.	A match.json replays to the same hash.
L2 Knife-fight	7-hex map. Phone pads or hotseat-with-pass.	Three humans groan once and want another.
L3 Room	Dumb WS room: lock → resolve → cinema on a shared view.	Strangers finish a game you are not in.
L4 World map	28–32 territories, 5–6 regions.	Telemetry bands in range over 200 bot games + 5 human.

Stack: TypeScript strict, packages/engine with zero UI imports, Vite + React + SVG client, Vitest, mulberry32 or equivalent. No Math.random in the engine.

03 — RULES LOCK

3.1 Players, map, setup, victory

- Players: 3–5. Design center is 4. Two-player uses a neutral third stack (section 3.9).
- MVP map: 7 territories, 2 regions, one choke. This ships first.
- Full map: 28–32 territories, 5–6 regions. Not 42. Interiors must be able to matter in a short game.
- **Deal:** shuffle territory ids, deal as evenly as possible. Leftover territories (when 42-style counts do not divide) become *unoccupied* (owner null, armies 0). They are legal attack destinations.
- **Starting armies:** each player gets a pool by player count (3p/35, 4p/30, 5p/25 as a starting point). Placement is simultaneous and capped. If a player cannot legally place their full pool, they place to cap and leftover armies are forfeit. After a scattered deal, run a second simultaneous placement pass before round 1 rather than inventing an exemption.

• **Victory (default):** after 8 rounds on the full map (5 on the knife-fight), highest score wins. Score = territories + region bonuses currently held. Ties: most regions, then most armies.

• **Early end:** control ≥ 80% of territories at upkeep, or last player with a territory.

• **Elimination:** 0 territories at upkeep. Packets already in transit still resolve this round. Cards from the eliminated player go to the owner of the packet that removed their last territory (queue order). If that packet later loses spoils, the cards already transferred stay with the eliminator.

3.2 The sealed sheet (one commit per player per round)

v0.1 put tactical after resolution in the same round. That forces a second handoff and makes resolveRound take orders that do not exist yet. v0.2 folds regroup into the next sheet.

On the sheet	Secret?	Notes
Regroup	Secret	Optional. From one friendly hex to one adjacent friendly. Leave ≥1 if origin should stay yours. Uses the map as it exists at the start of this planning window (post last resolve).
Cash set	Secret	Optional. One set. Armies enter the income pool this sheet.
Place income	Secret	Income computed on the post-regroup map. Cap applies. Overflow lost.
Attacks	Secret	0–N packets. Surge is a second destination on a packet, not a flag.

Round 1 has no regroup. After sheets lock: apply all regroups simultaneously → apply all placements → vacate all packets → run the battle queue → upkeep (cards, elimination, score, victory). Then the next planning window opens on the new map. One device pass. One lock.

Just-placed armies may attack this round. That is implied by the sheet order and is now explicit.

3.3 Income

Income = max(3, floor(territories / 3)) + floor(largestConnectedBloc / 6) + regionBonus + cashedSet.

v0.1 added owned + connected at full weight and doubled classic income for a blob. Connected still matters; it is no longer a second copy of every hex. Region bonuses: small 2, mid 3–4, large 5. Recalibrate only against telemetry, not against a feeling.

Placement cap unchanged: max added to T = 1 + number of T's adjacent hexes you own *after regroup, before place*.

3.4 Packets

A packet is `{from, to, strength, surgeTo?: TerrId}`. Legal when: from is yours; to is adjacent and not yours (or is unoccupied); strength ≥ 1; after summing all outgoing strength from from, at least 1 army remains on from; if surgeTo is set, it is adjacent to to and not equal to from.

Same-owner merge: before the queue, all packets from the same player with the same to combine strength and keep a surgeTo only if every merged packet named the same surgeTo (otherwise surgeTo is dropped). Different owners never merge.

On lock, strength leaves from. Those armies are in transit. They do not defend from. They do not defend to until they win it.

3.5 Combat — v0.1 revoked

v0.1 used stacked attacker edges plus loser-loses-2 volleys. Measured: 1v1 mutual wipe 100%, 8v5 capture 99%, 4v4 a 46% double wipe, 5v5 a different game because of a mod-3 artifact. That combat cannot support the vacate rule. It is out.

Rule (one roll, one thumb, size in the open):

• Each side rolls **1d6 and adds its army count**.

• Higher total wins. **Ties go to the attacker.** That is the only attacker advantage.

• The loser is removed.

• The winner loses floor(loser / 2) armies, but always keeps at least 1 if they won.

• A 0–0 result therefore cannot happen. 1v1: winner occupies with 1.

• Clash and spoils have no defender. Both sides are attackers. **Reroll ties.** Winner-loss rule still uses the other packet's pre-roll size.

• Dice counts do not change mid-fight. There is no mid-fight 2× bonus. There is no volley loop.

Simulated capture bands for this rule (8k+ fights, ties → attacker):

Matchup	Capture	Band we want	Notes
1v1	58%	55–60 even	Winner keeps 1. Remnant can spike.
3v1	83%	remnant taxes	1-man doorstop is real, not a hero.
even any size	58%	55–60	Stable at 4v4, 8v8, 20v20. No compound grind.
6v4 (3:2)	83%	~80	Favored, not free.
8v5	91%	high 80s / low 90s	Acceptable. Bloody occupation.
5v8	16%	≤ 25	Defense of a real stack works.
10v5 (2:1)	100%	~95	Half-size garrisons die. Fine.

UI may show a band on the order sheet (long shot / toss-up / likely / lock). Do not print 91.2%. The groan is the product, not the spreadsheet.

3.6 Queue

• **A Border clashes.** If A sent a packet into B's hex and B sent a packet into A's hex, those two packets fight each other first (both attackers, reroll ties). Survivors continue as invasions into the opposing hex with remaining strength. The 1 left at each origin is not in this fight.

• **B Invasions.** Every remaining packet vs the current garrison of its destination. If several *different owners* hit the same dest, they fight the garrison one at a time in seeded order. A packet that wipes the garrison occupies immediately and becomes the garrison for the next packet.

• **C Spoils.** If two or more different-owner packets still have transit armies and the dest is unoccupied or was just emptied without a sitting occupier, they fight as attackers (reroll ties) until one packet remains. That packet occupies.

• **D Surge.** After A–C, a surviving occupier that declared surgeTo and has ≥ 2 armies may send (all but 1) at surgeTo. Two surges that land on the same hex are a spoils fight. A surge at a hex you already own is cancelled. A surge at a hex taken by someone else this queue hits the new owner.

• Unoccupied hexes (owner null, 0) are won by the first packet that reaches them in queue order; further packets spoils-fight that occupier.

3.7 Cards

• Deck: 3 types × N, plus 2 wilds. N = number of territories on the map (knife-fight can use a tiny deck or skip cards entirely — skip them on the 7-map).

• Capture ≥ 1 territory this resolve → draw 1 at upkeep. Hand 5. At 5 you must cash on the next sheet before you can draw again.

• Set: three matching or one of each (wild substitutes). Table escalation: 4, 6, 8, 10, then +5. Global counter, not per player.

• No other card powers in MVP.

3.8 Unoccupied and 0-army hexes

A hex is either owned with ≥ 1 army, or unoccupied (owner null, armies 0). There is no owned-empty state. If a rule would create one, the hex becomes unoccupied. Walking into unoccupied still requires a packet of ≥ 1 and still vacates the origin — taking a hole in the line is not free.

3.9 Two players

A pure 1v1 simultaneous duel is matching pennies. MVP two-player uses a Neutral: after the deal, leftover hexes plus a fixed extra pile are Neutral. Neutral does not plan. At resolve it only defends with the size+1d6 rule. Neutral never attacks. Region bonuses Neutral holds are unpaid. This is a doorstop, not an AI opponent.

3.10 Planning clock (digital)

Knife-fight: 90 seconds after the first lock-ready or a generous first-round 3 minutes. Full map: 2 minutes. On timeout, the engine accepts the last valid partial sheet or a hold sheet (no regroup, no place, no attacks). Disconnect: same as timeout after a short grace. Do not stall the table.

04 — ENGINE CONTRACT

Split static map from live state. Do not grow a log onto World and then hash World.

```
export type PlayerId = string
export type TerrId = string

export interface MapDef {
  id: string
  territories: Record<TerrId, { name: string; region: string; adjacent: TerrId[] }>
  regionBonus: Record<string, number>
}

export interface World {
  mapId: string
  round: number
  territories: Record<TerrId, { owner: PlayerId | null; armies: number }>
  players: Record<PlayerId, { name: string; color: string; cards: Card[]; alive: boolean }>
  setsCashed: number
}

export interface Packet {
  from: TerrId; to: TerrId; strength: number; surgeTo?: TerrId
}
export interface Sheet {
  playerId: PlayerId
  regroup?: { from: TerrId; to: TerrId; strength: number }
  cashSet?: Card[]
  placements: Record<TerrId, number>
  packets: Packet[]
}

// seed is NOT stored on World. Server / CLI holds matchSeed.
resolveRound(world, map, sheets, matchSeed) → { world, events: Event[] }

validateSheet(world, map, sheet) → Error[]
incomeOf(world, map, playerId) → number
capOf(world, map, terrId) → number
connectedBloc(world, map, playerId) → number

// Replay = { mapId, matchSeed, setup, sheets: Sheet[][] }
// Hash for T6 = hash(world snapshot AFTER resolve), never hash(seed + world).
```

Event is a required type, not a later nicety. Minimum kinds: RegroupApplied, Placed, Vacated, Clash, Invasion, Spoils, Surge, Occupied, BecameUnoccupied, CardDrawn, PlayerEliminated, Score, RoundEnded. Each event names the hexes, the players, the pre/post sizes, and the two dice + totals when combat happened. Cinema, replay, and “no hidden engine” all render this list.

RNG: derive a substream per battle from hash(matchSeed, round, battleKey). Adding an unrelated packet on the other side of the map must not reroll this fight. matchSeed stays off the public World until the round is locked and resolved. Hotseat can be honorable. A net client must not be able to pre-sim the dice.

05 — REPO

```
concurrent/
  packages/engine/src/{types,map,income,validate,combat,resolve,rng,events}.ts
  packages/engine/test/{combat,resolve-order,income,properties,fixtures}/
  tools/combat_sim.ts          # band check against section 3.5
  packages/maps/{knife7,world30}.ts
  apps/cli/                    # text map, JSON sheets, replay
  apps/web/                    # SVG map, sheet UI, cinema, ghost arrows
  apps/room/                   # later: lockstep WS, still calls resolveRound
  docs/CONCURRENT_Build_Bible_v0.2.pdf
  README.md                    # one command to knife-fight
```

06 — SEQUENCE (CLOCKS DROPPED, GATES KEPT)

Gate	You may not proceed until
G0 Scaffold	Workspaces, strict TS, rng with no Math.random, types compiled.
G1 Combat	combat_sim prints bands inside section 3.5. 1v1 never double-wipes.
G2 Queue	T1–T8 green on the 7-map fixtures, including three-way clash→mass→spoils.
G3 Monkey	Random-legal bot plays 1,000 games. Properties: no negatives, army accounting, every hex owned or unoccupied, determinism.
G4 CLI	You can play a knife-fight against the monkey and dump match.json.
G5 Visible	Ugly client: sheet + map + cinema + ghost arrows of last round. Cap readout on enemy hexes.
G6 Human	Three humans finish. Vacate-punish happens. They ask for another.
G7 Room	Phones submit sheets at the same time at a table with one shared cinema.
G8 World	28–32 map only after G6. Tune income/regions against bot telemetry + those humans.

A random-legal bot is not week-6 content. It is how you know resolve.ts does not leak armies. A greedy planner can wait. Factions, paper port, diplomacy-as-a-phase, fortification markers: out of this bible until G6 is true.

07 — TESTS

- **T1 Vacate-punish.** Red origin 5 sends 4. Green sends 3 at that origin. After lock Red origin has 1. If Green wins, Green owns it even if Red's packet later wins elsewhere.
- **T2 Mutual strike.** Clash packets fight each other before either leftover garrison is used as a defender.
- **T3 Mass then spoils.** Red 5 and Blue 5 both hit Green 3. One owner remains.
- **T4 Illegal leftover.** Origin 3, two packets of 2. Rejected.
- **T5 Cap.** Isolated hex accepts at most 1 reinforcement. Overflow vanishes.
- **T6 Determinism.** Same setup + matchSeed + sheets ⇒ same world hash and same event list.
- **T7 Surge last.** Surge target that is also a primary is resolved as a primary first.
- **T8 Three-way.** Clash survivor continues into a hex that a third player also hit. Log is reconstructible from events.
- **P1–P4 Properties.** Fuzz random legal sheets: no negative armies, Σ armies + in-transit accounted, no owned-empty hex, validateSheet rejects what resolve would choke on.

08 — UI THAT SERVES THE WHEEL

- Map is a status board. Counts readable. Vacated origins look vacated the moment of lock, before dice.
- Enemy hexes show current armies and max possible after their cap. Hidden reinforce is bounded, not fog.
- Sheet lists each packet with origin-remaining in red if it will sit at 1.
- Ghost arrows: last round's packets stay on the map at 30% while you plan this round. Teaching tool, now required.

- Cinema steps the queue. Pause. One event per beat. The log is the same Event[] the engine emitted.
- Odds as words, not percents.
- Colorblind patterns on owners. Lock is irreversible for that round.

09 — TUNE VS LEAVE

Knob	Touch?	Why
Vacate	No	The game.
Tie → attacker (invasion only)	No	The one thumb. Clash/spoils already reroll.
Size + 1d6	Only if bands drift	Do not restack edges.
Winner loss = floor(loser/2)	Yes, second	Bloodiness lever.
Income and region values	Yes	Snowball lever.
Cap formula	Rarely	It is doing hidden-info work.
Round count / 80% early end	Yes	Length lever.
Surge exists and goes last	No	Breakthrough reward, no pre-empt.
Cards on 7-map	Skip	Noise until the wheel is visible.
Factions / missions / paper	After G6	Not how this becomes 2026.

10 — TELEMETRY (OR YOU ARE NOT TUNING)

Log per match, from the monkey and from humans:

Metric	First target
Rounds to end (full map)	6–8 median
Vacate-punish frequency	≥ 1 event in ≥ 40% of human games
Lead changes after round 3	> 0 in most games
Attacker capture at even sizes	55–60%
Attacker capture at 3:2	75–85%
Games a human wants to replay immediately	qualitative gate G6

11 — DEFINITION OF DONE, v0.2 PLAYABLE

- G6 is true on the knife-fight map.
- combat_sim is in the repo and still prints bands inside the table in 3.5.
- A match.json plus matchSeed replays to the same world hash.
- No Math.random in packages/engine. Seed is not on public World pre-lock.
- README starts the client in one command.
- You did not add a hook to justify the project. The walk-in on a vacated origin is the hook.

SameTime was already the idea. Build the kernel that can run it in the open, on one lock per round, with dice a person can see. Everything else waits.

v0.2 amends v0.1. Combat 3.5 of v0.1 is not a source. Next amendment needs a failing test or a band that moved.